Tempora-Fusion Artifact Evaluation Manual

A Artifact Appendix

A.1 Abstract

This artifact is a C++ research prototype of Tempora-Fusion, the Verifiable Homomorphic Linear Combination Time-Locked Puzzle (VHLC-TLP) construction in Section 5.2 of the paper. It contains protocol and primitive translation units, their required project headers, a test harness, and the top-level `README.md`. The harness generates synthetic cryptographic inputs, creates ten client puzzles, computes an equal-weight combination using three leaders, solves the combined puzzle and client puzzle 0, verifies both, and prints correctness checks and phase timings. There is no dataset. Version: **1.0**. License: **MIT**.

A.2 Description & Requirements

`README.md` is the evaluation entry point and contains the complete source inventory, dependency commands, build command, expected-output checklist, parameter table, limitations, and troubleshooting guide. `TestRoutines.cpp` contains `main()` and the fixed experiment configuration. The orchestration modules are `setup.cpp`, `gen_puzzle.cpp`, `linear_comb.cpp`, `solve_puzzles.cpp`, and `verify.cpp`; the remaining modules provide RSA/TLP, AES-based PRF, BLAKE3 commitments, coin tossing, OT, OLE/OLE+, and finite-field interpolation. All project headers referenced by the translation units must be included; `TestRoutines.cpp` is the entry point and has no same-named header. Files renamed with download suffixes such as `(3)` must be restored to their canonical names.

A.2.1 Security, privacy, and ethical concerns. Execution uses synthetic random data, is offline, requires no privileged access, and neither reads nor transmits personal data. Dependency installation may modify a system prefix if run with `sudo`; an unprivileged prefix is acceptable. The program is CPU-intensive and the OT prototype retains some allocations until process exit. Use the documented 30-minute timeout. This is prototype code and must not be used to protect production secrets.

A.2.2 How to access. The permanently archived reviewed release is available at https://anonymous.4open.science/r/Tempora-Fusion-1F05/README.md.

A.2.3 Hardware dependencies. None beyond a commodity 64-bit Unix-like machine. No GPU, enclave, or network service is required. We recommend at least one CPU core, 4 GB RAM, and 1 GB free disk. Runtime is CPU- and library-build-dependent. However, the compute time as listed by the workflow will largely depend on the hardware *and* the number of leader/non-leader clients used in the experiment (Section A.5).

A.2.4 Software dependencies. A C++17 compiler, standard build tools, NTL, GMP, and `cryptopp-modern` are required. Upstream Crypto++ 8.9 is insufficient because `commitment.cpp` uses BLAKE3. On AArch64 use `cryptopp-modern` 2026.5.1 or later. Exact tested versions must be recorded in the archived `README.md`. Installation references are https://github.com/cryptopp-modern/cryptopp-modern, https://libntl.org/, and https://gmplib.org/.

A.2.5 Benchmarks. None. The artifact generates ten random 32-bit messages and all cryptographic material at run time. The default is a functional experiment, not a statistically controlled benchmark or a calibrated real-time-delay study.

A.3 Set Up

A.3.1 Installation. The following section may be automatically done with a provided script (`install_script.sh`). **Run inside the directory with all repository files.**

On Ubuntu, install `build-essential`, `ninja-build`, `cmake`, `libntl-dev`, and `libgmp-dev`; then build and install `cryptopp-modern` following its official CMake instructions. No artifact-specific installation script is required. From the artifact root, run the single `g++` command in `README.md`. It creates or overwrites only `VHLC_TLP.exe`; compilation must exit with status 0. Do not define `NDEBUG`, because the implementation uses assertions.

A.3.2 Basic test. Run:

```
timeout 30m ./VHLC_TLP.exe 2>&1 | tee run.log
sed -r 's/\x1B\[[0-9;]*[mK]//g' run.log > run.clean.log
```

The run is offline and may be repeated safely. It creates only the two named logs. Success requires exit status 0, the OT line `PASS: All 1000 OT executions correct`, and the detailed conditions in `README.md`; the final `All tests passed.` line alone is not a sufficient oracle.

A.4 Evaluation Workflow

A.4.1 Major claims.

- (C1): The artifact exercises the complete Section 5.2 implementation path: setup, client-puzzle generation, homomorphic evaluation, solving, and verification. This is supported by experiment (E1).
- (C2): For the generated ten-client, three-leader, unit-coefficient instance, the solved evaluated puzzle equals the direct field sum of the ten plaintext messages, and the emitted root/commitment checks accept. This is supported by experiment (E2).
- (C3): The prototype reports wall-clock timings for major protocol phases. Experiment (E3) makes these observations available, but they are single-run measurements rather than a paper-level comparative benchmark.

The artifact does not reproduce formal proofs, calibrated delay claims, comparisons with other systems, or unidentified paper tables/figures. It has no fixed seed, repeated-trial driver, structured-data exporter, or plotting script. Quantitative paper claims require additional claim-to-result cross-references and data before submission for *Results Reproduced.*

A.4.2 Experiments.

(E1): **Build and smoke test** [15 person-minutes + < 2 compute-minutes, excluding dependencies].
Preparation: Confirm canonical filenames, all headers, and dependency versions as specified in `README.md`.
Execution: Compile with the documented C++17 command and launch the executable with the 30-minute timeout.
Results: Require compiler and process status 0 and the OT pass line. Missing headers indicate an incomplete archive; missing BLAKE3 indicates the wrong Crypto++ distribution.

(E2): **End-to-end functional validation** [5 person-minutes + up to 30 compute-minutes].
Preparation: Retain `run.clean.log` from (E1).
Execution: Extract lines matching `True sum`, `PzlEval`, `Commitment check`, `Validity check`, and `Verification for`.
Results: Require equality of `True sum` and `PzlEval`; every emitted commitment/root check must say `Pass`; both puzzle decisions must say `Accepted`. Values and leaders vary across runs. The harness does not assert verifier return values, so all explicit checks are mandatory.

(E3): **Timing inspection** [5 person-minutes; no additional compute time].
Preparation: Use the successful (E2) log and record CPU, OS, compiler, optimization flags, memory, and dependency versions.
Execution: Extract all lines ending in `execution time:`.
Results: Report the raw log and environment. Interpret timings only as one random default run; no fixed numerical tolerance is claimed.

A.5 Notes on Reusability

Alternative client/leader counts, coefficients, message sizes, and delay parameters can be set in `testVHLCTLP()` as described in `README.md`; rebuild after each change. Reusers must preserve the field and security-parameter preconditions and report all modifications.